EENG 385 - Electronic Devices and Circuits
Full Bridge Rectifier With Buck Converter
Lab Handout

Outcome and Objectives

The outcome of this lab is to build a full-bridge rectifier to convert an AC input to the DC voltage. Through this process you will achieve the following learning objectives:

- Analyze and design a circuit containing one or more diodes.
- Use laboratory test and measurement equipment to analyze electronic circuits.

Full-bridge Rectifier Theory:

A rectifier is an electrical device that converts an AC signal into a DC signal. A full-bridge rectifier is a rectifier that uses the arrangements of four diodes shown in Figure 1 along with the 100uF capacitor C1. The input to this rectifier is the 17V AC signal on the left side of Figure 1. The DC output of the rectifier is the voltage difference Vrec+ to Vrec-. The load is modeled as the 100Ω resistor R1.

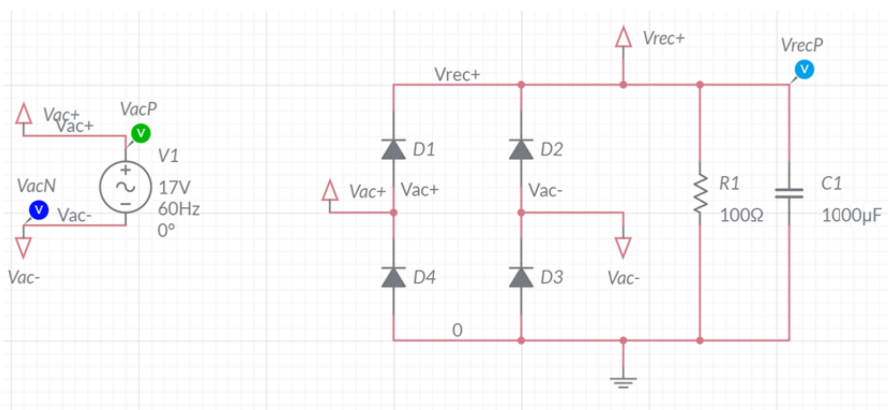


Figure 1: The full bridge rectifier converts the 17V AC input into DC output.

Let's start our journey by examine the function of the four diodes in Figure 1 by applying the waveform shown in Figure 2 as the AC input. For this thought experiment, remove the resistor R1 and capacitor C1 from the circuit. The green waveform in Figure 2 is Vac+ and the blue waveform is Vac-. Assume a diode drops 0.7V when forward biased.

1) If Vac- is less than -0.7V with respect to GND, what diode is conducting?
2) If Vac+ is 0.7V higher than Vrec+, what diode is conducting?
3) When Vac- is below -0.7V and Vac+ is above 0.7V, what pair of diodes are forward biased (conducting) and what pair are reverse biased (off).

4) When Vac- is less than -0.7V, the Vac- terminal of the AC source is fixed at -0.7V. Thus, during this time the amplitude of Vac+ is referenced to -0.7V. What is the peak amplitude of Vrec+ during this time?
5) If Vac+ is less than -0.7V with respect to GND, what diode is conducting?
6) If Vac- is 0.7V higher than Vrec+, what diode is conducting?
7) When Vac+ is below -0.7V and Vac- is above 0.7V, what pair of diodes are forward biased (conducting) and what pair are reverse biased (off).
8) When Vac+ is less than -0.7V, the Vac+ terminal of the AC source is fixed at -0.7V. Thus, during this time the amplitude of Vac- is referenced to -0.7V. What is the peak amplitude of Vrec+ during this time?

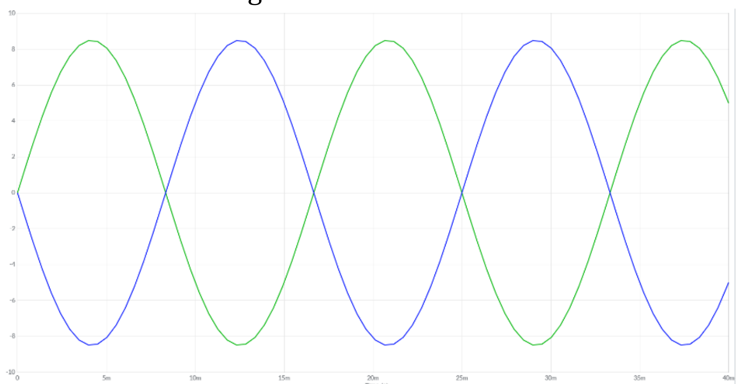


Figure 2: The Vac+, Vac- input to the full-bridge rectifier.

Now that you have a better understanding of the role of the 4 diodes in Figure 1, let's explore the role of the capacitor and resistor.

The addition of the capacitor (but no resistor) to the rectifier acts to hold the highest value of Vrec+ on the positive plate of the capacitor. The addition of the load resistor causes the stored charge on the capacitor to bleed off through the resistor when the input waveform is less than Vrec+.

The capacitor C1 in Figure 1 is called the "filter capacitor" because, along with the load resistor, forms a low pass filter with time constant R1*C1. If the time constant of this low pass filter is large compared to the period of the 60Hz ac input, the capacitor voltage will look like a DC voltage with low amplitude 60Hz ripples between the peaks of the AC input. We will explore these ripples in the following section.

Full-bridge Rectifier Simulation:

Let's look at the detailed behavior of the full bridge rectifier by simulating the circuit in Figure 1. Do this by logging into your Multisim Live account. I've setup the simulation today because you have a lot of work to complete, so just open this link and you should see the assembled circuit similar to Figure 1. Run your simulation, it should stop after 40ms and your Grapher output should look like Figure 3.

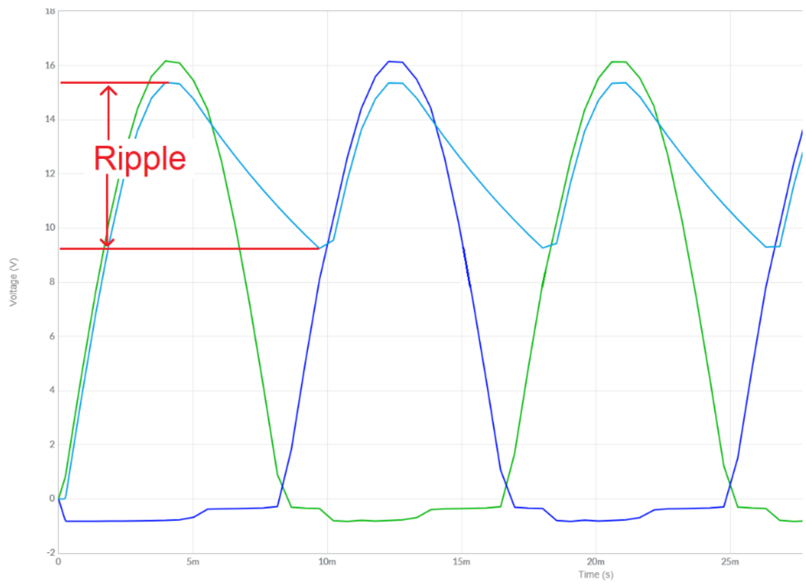


Figure 3: Ripple voltage is the difference between the highest and lowest voltage on Vrec+. The green and dark blue traces are the Vac+/Vac- input respectively, the light blue trace is the Vrec+ output.

One attribute of a power supply is the ripple voltage which is the peak-to-peak variation of the output. In Figure 3, the ripple voltage is 15.4V – 9.3V = 6.1V. The duration of this drop in voltage starts at the peak of Vac voltage and ends when Vrec+ starts increasing. In Figure 3, the duration of the voltage drop is 18ms – 12.3ms = 5.7ms.

The ripple voltage depends on the resistor and capacitor in Figure 1 because once the Vac voltage starts dropping off, the charging diodes become reverse biased, leaving the stored charge on the filter capacitor C1 to supply the output voltage. Since this capacitor voltage is in parallel with the load resistor R1, the capacitor discharges. This is an RC decay with the familiar equation, $V_0e^{-t/R1*C1}$

Let's take a moment to measure the ripple voltage with different loads and compare it against a theoretical RC decay model in Table 1. To do this, simulate the full-bridge rectifier for the load resistors shown in the "Load Resistor" column in Table 1.

- Use the Grapher output to measure the ripple voltage (as shown in Figure 3). Put this value in the "V_{sim_ripple}" column. Round to 3 significant figures.
- Multiply the load resistor times the filter capacitor and put this value in the "RC" column.
- Calculated the power dissipated through the load resistor using the $V_{average}{}^2/R$ equation. Do this by assuming that that $V_{average}$ = 15V and R is the load resistor. Put this value in the Power column using the units of Watts. Round to 2 significant figures.

Table 1: The ripple voltage depends on the load resistances.

Load Resistor	V_{sim_ripple}	RC (ms)	Power (W)	V_{act_ripple}
100Ω				
470Ω				

Full-bridge Rectifier Implementation:

The input to the Power Converter PCB comes from an 120VAC to 17VAC transformer that you will plug into a wall outlet. To insure that you use this device safely, let's explore how it works.

The KS21239L4 Transformer

Details on the KS21239L4 transformer are hard to come by (non-existent), but we can determine all the relevant details of the transformer using educated guesses and making measurements with your test and measurement equipment. **You will start by making measurements with the transformer unplugged and with all stray wires removed from the screw terminals!** You can get a screwdriver from the tool cabinet in Brown 304 to remove all wires connected to the screw terminals.

Before you proceed, grab a pair of digital multimeter (DMM) cables, one red and one black. Connect the black cable to the COM jack on the BK Precisions DMM. Connect the red cable to the VΩ jack. Now, pick up and inspect the KS21239L4 transformer. The physical interface consists of a 3-prong 120VAC electrical connection and 4-screw terminal labeled 1, 2, 3, 4/GRD. I would bet that the GND prong of the 120VAC connector is connected to the screw terminal labeled "4/GRD". Test this hypothesis by setting the BK Precisions DMM in the resistance mode (press the Ω button) and measuring the resistance between the GND prong and the "4/GRD" screw terminal.

1) What is the resistance between the GND prong and the "4/GRD" screw terminal? Are these two electrically connected?

Now that we have that out of the way, let's determine the configuration of windings in the transformer. My guess is that the transformer output is center tapped; meaning that the secondary winding is divided into two evenly split sections as shown in Figure 4.

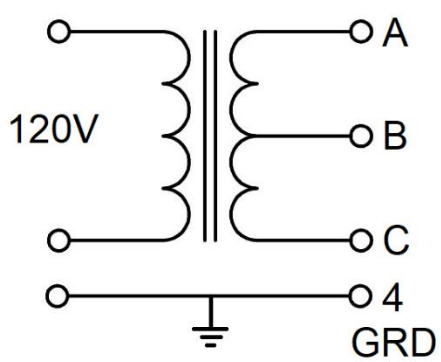


Figure 4: An educated guess at the internal organization of the KS21239L4 transformer.

In the following, we will assume that all the wire used in the secondary winding has the same resistance per unit length. If this is the case, then:

- The resistance between A and B will equal the resistance between B and C.
- The resistance between A and B will be half the resistance between A and C.

Use the DMM to probe the screw-terminals 1, 2 and 3. Note, the resistance between A/B and B/C may not be exactly equal, use an average value in your answer below.

2) What is the resistance between the A and B terminals?
3) What is the resistance between the A and C terminals?
4) What screw-terminals are associated with A, B and C in Figure 4? Hint, there are two possible answers, just provide one.

The final answer that we need, is what screw-terminals we should use as the input to our transformer? To do this, you will need to plug in the transformer while having access to the screw terminals. I found that an extension cord works the bet and surge suppressor can make due.

The case of the transformer has printed text that described the voltage and current ratings of the primary and secondary.

5) What is the voltage for the primary winding and secondary winding?

The voltage specifications for the transformer are root mean squared (RMS), whereas and measurements that you will make on the oscilloscope are peak-to-peak. The relationship between the two is:

RMS voltage * sqrt(2) = Peak-to-peak voltage

Before proceeding, put the DMM into AC voltage measurement mode by pressing the **AC V** button. Now, plug in the transformer and use the DMM to measure the voltages of the following screw terminals.

6) What is the RMS voltage between terminals A and B in Figure 4? Convert this RMS voltage to peak-to-peak.
7) What is the RMS voltage between terminals A and C in Figure 4? Convert this RMS voltage to peak-to-peak.

Now attach a pair of wires to a pair of screw-terminals whose peak-to-peak voltage is close to 17V, the voltage that we used in the simulations of the previous section. These two wires will provide Vac+ and Vac- to your circuit.

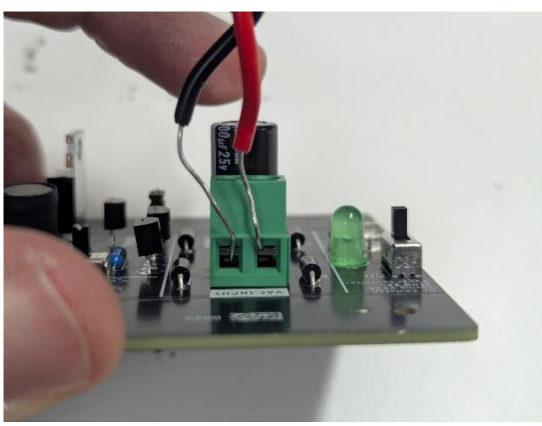


Figure 5: When connecting the transformer to the Power Conversion PCB, make sure to put the wire above the gate (red wire), not below the gate (black wire). The black wire will not make a reliable electrical connection and will fall out.

The Power Conversion PCB

The Power Conversion PCB is shown in Figure 6. The ON/OFF switch is on the left side. The PWM signal for the buck converter connects to the PWM/GND terminals on the left side. The transformer is connected to the green VAC INPUT screw terminal. The pair of switches, S1 and S2, at the top select where the load is applied and the magnitude of the load. You can connect the GND clip of an oscilloscope probe to the GND terminal on the upper right and probe the adjacent signals.

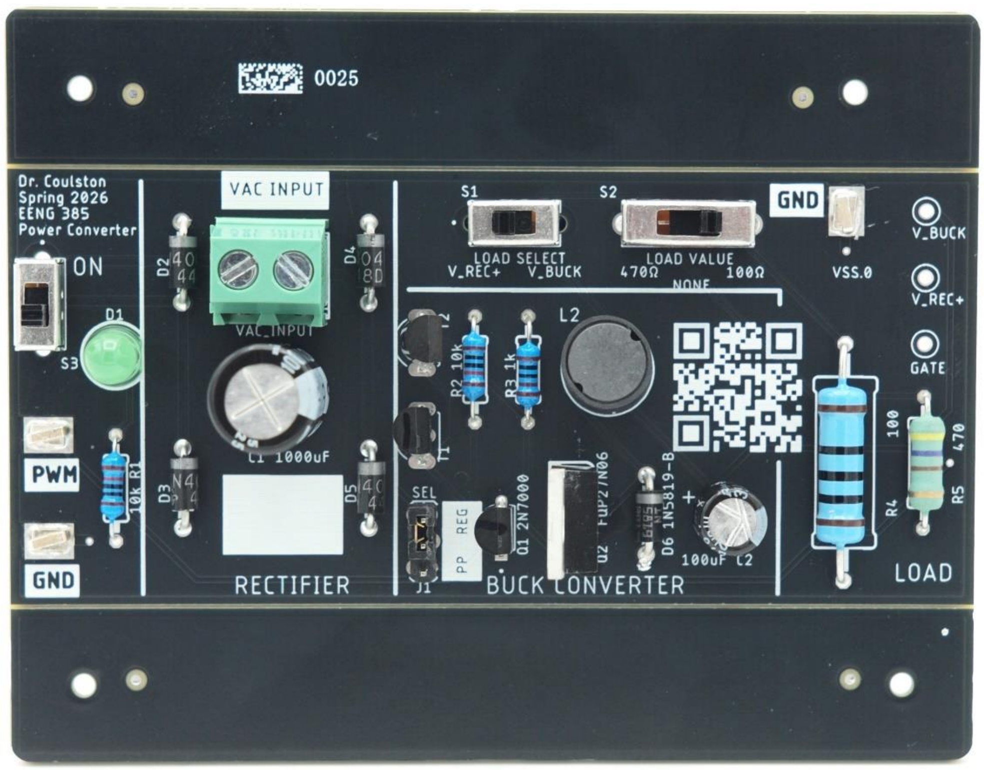


Figure 6: The Power Conversion PCB.

The Power Conversion PCB has two major subsystems, the full-bridge rectifier and the buck converter. You will be working with the full-bridge rectifier subsystem whose schematic is shown in Figure 7. Let's examine this schematic so you understand how the Power Conversion PCB operates.

The AC input from the KS21239L4 transformer enters the board through the pair of connectors in the upper left corner of in Figure 7. This AC input is connected to switch S3, an ON/OFF switch, that illuminates an LED when the PCB is powered on. The rectifier is connected to diodes D2, D3, D4, D5 and the 1000uF capacitor C1. You can probe the rectified voltage at the testpoint V_REC+ shown on the left side of the schematic. Switch S2 allows you to connect the rectified voltage to one of three loads (100Ω, 470Ω or no-load).

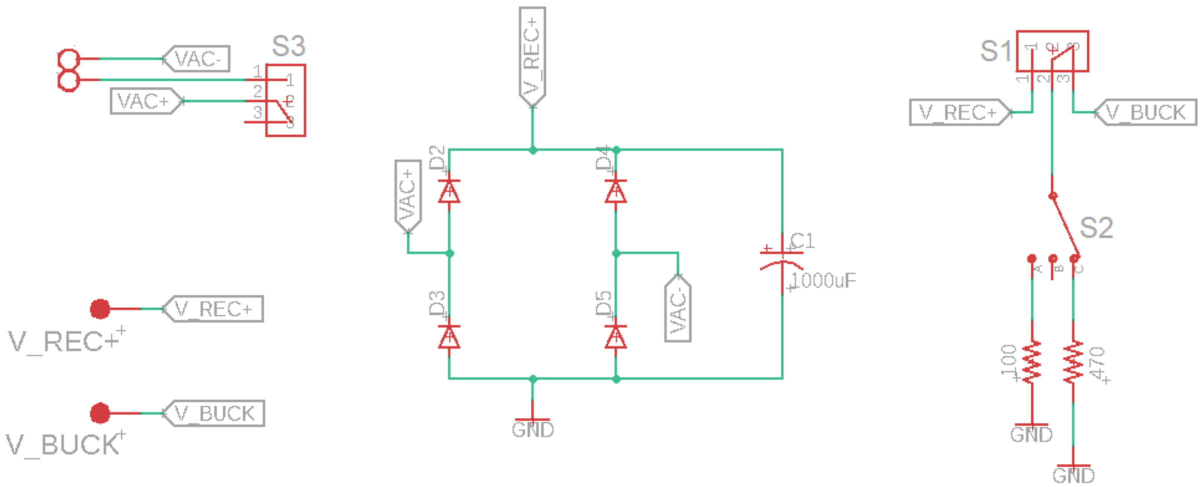


Figure 7: The schematic for the rectifier portion of the Power Conversion PCB.

Now that you understand what the Power Conversion PCB is doing, let's set it up to determine the effect of resistance on the rectified voltage. You will put your finding into the V_{act_ripple} column of Table 1. Start by setting up the Power Conversion PCB as shown in Figure 8.

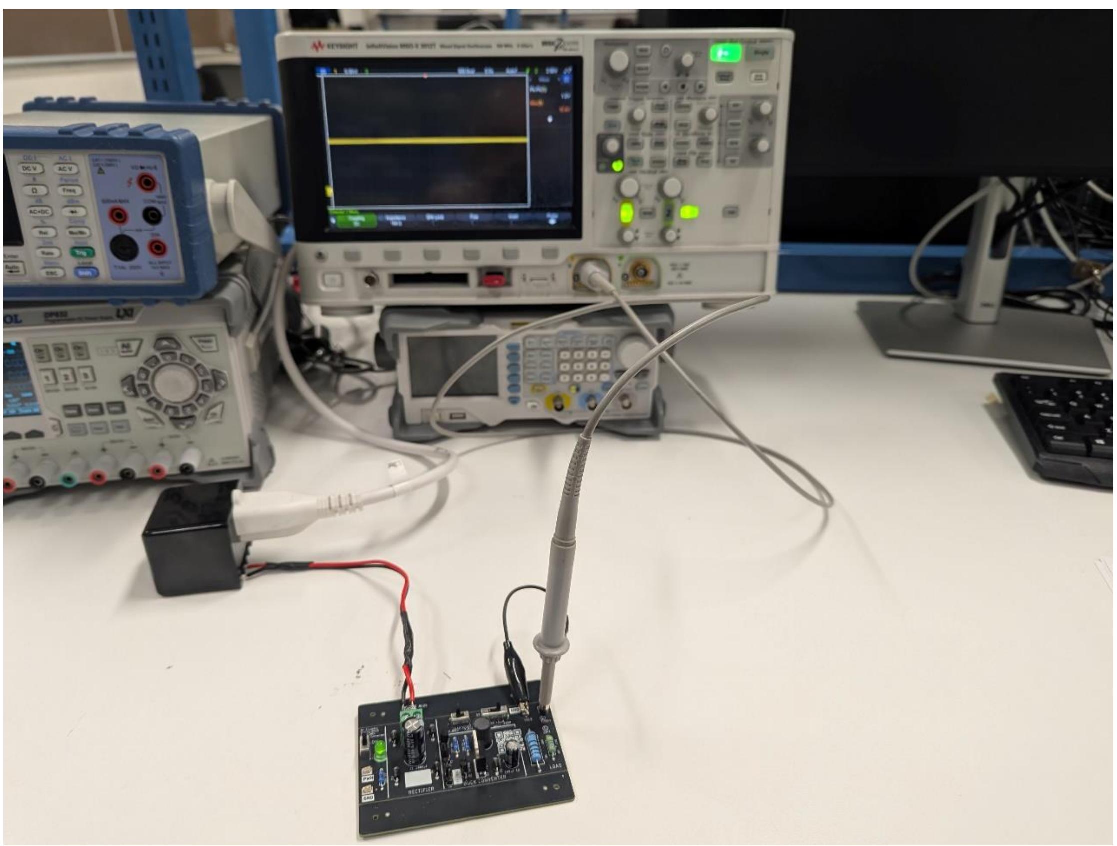


Figure 8: The Power Conversion PCB setup to make measurements for V_{act_ripple} column in Table 1..

Next, configure your oscilloscope as follows:

Horizontal (scale)	5ms
Ch1 probe	V_REC+ terminal
Ch1 (scale)	5V/div
Ch1 (coupling)	DC and AC
Trigger source	1
Trigger slope	↑
Trigger level	10V
LOAD SELECT	V_REC+
LOAD VALUE	None
Measurements	PK-PK (channel 1) MAX (channel 1)

Your oscilloscope output looks like Figure 9.

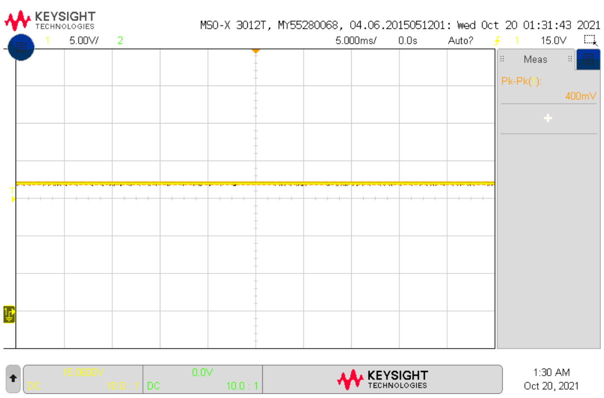


Figure 9: The V_REC+ output of the full-bridge rectifier with no load resistor in the circuit.

Now move switch S2 into the 100Ω setting to measure the ripple voltage. Your oscilloscope should look similar to Figure 10. In order to get the best measurement, switch channel 1 to AC coupling and increase the vertical magnification on channel 1. <u>Make sure to grab a screen shot of the waveform for your lab report.</u>

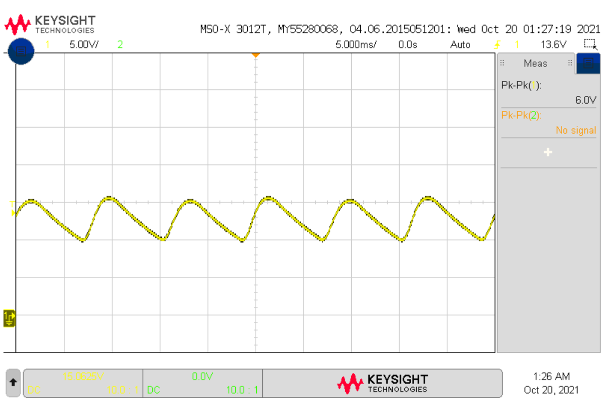


Figure 10: The V_REC+ output of the full-bridge rectifier with a 100Ω load resistor.

Now measure the ripple of the Vrec+ output for the 100Ω resistor. You might want to use the cursor function for this or one of the special measurement modes. Record this in the V_{act_ripple} column in Table 1. Next, switch a 470Ω into the rectifier load and record the ripple in the V_{act_ripple} column in Table 1 and grab a screen shot of the waveform for your lab report. While all these resistors are sized to handle the power dissipation, they will get hot to the touch, please be cautious when touching them.

Buck Converter Theory:

A buck converter is a DC/DC switching regulator that uses a diode, PMOS MOSFET, inductor and capacitor in a configuration like that shown (on the right side) in Figure 11. The voltage V_REC+ is the "output" from the full-bridge rectifier. **You should assume that V_REC+ is 15V DC**.

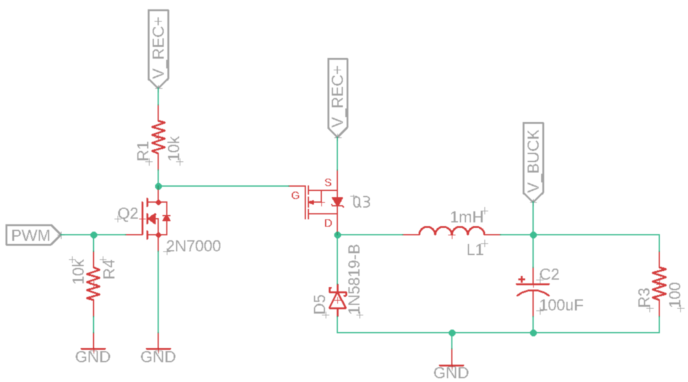


Figure 11: Schematic of a buck converter. The input is the PWM signal and the output is a DC voltage at Vout.

Let's explore how this circuit works by considering the MOSFETs Q2 and Q3 as switches. We do this by replacing the pair of MOSFET in Figure 11 with a switch that is either open or closed as shown in Figure 12. The state of each switch (whether it is open or closed), depends on the gate voltage. The gate of the MOSFET is horizontal signal incident to the switch's open/close paddle. For Q2 this is the PWM signal.

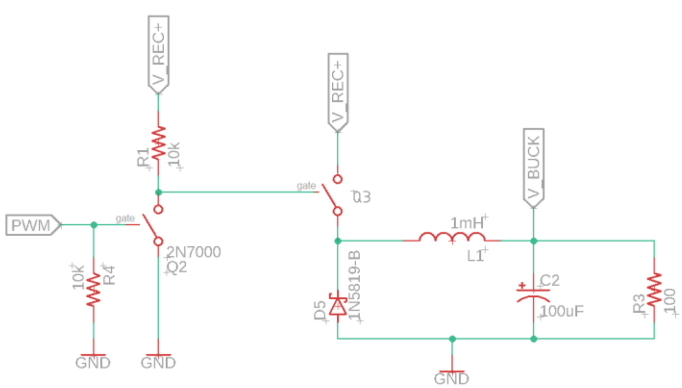


Figure 12: The buck converter with the MOSFETs replaced by switches whose position is controlled by the gate voltage.

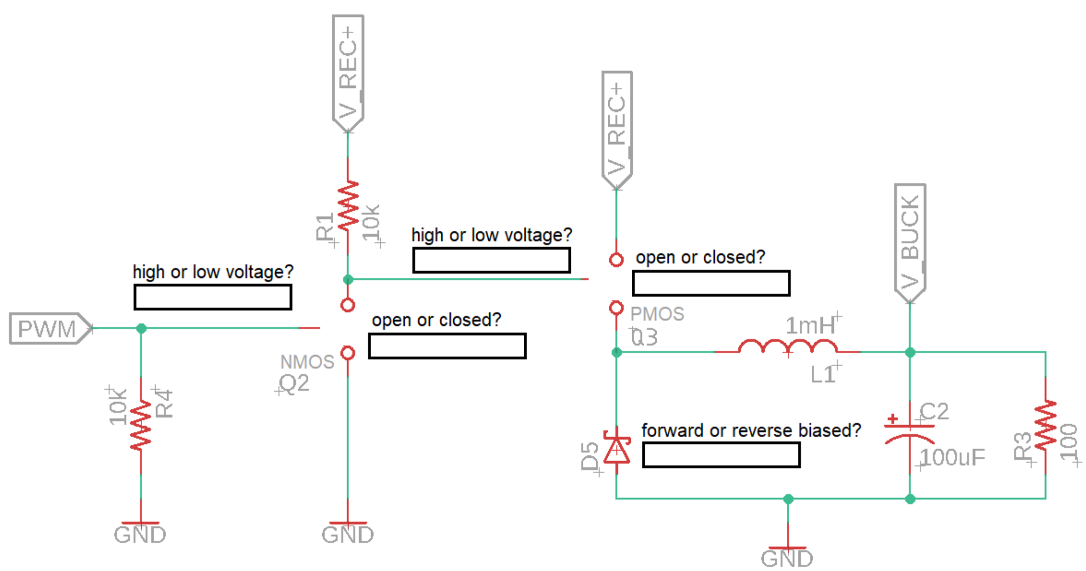
While it is difficult to tell from the schematic in Figure 11, the MOSFET Q2 is NMOS and Q3 is PMOS. NMOS and PMOS tell you how the circuit was manufactures, something you can learn lots more about in VLSI. For us, the difference is important because the rule governing the state of the switch is different for each and given in

Table 2: The state of a MOSFET switch as a function of the type and gate voltage.

Type\Gate voltage	LOW (ground)	HIGH (V_REC+)
NMOS (Q2)	OPEN	CLOSED
PMOS (Q3)	CLOSED	OPEN

Let's look at the circuit shown in Figure 12 when the **PWM signal is at a high voltage**. Let's start with Q2 because it is directly connected to the PWM signal.

1) What terminal of the Q2 MOSFET is connected to the PWM signal?
2) When the gate of Q2, a NMOS FET, is at a high voltage, is Q2 an open or closed switch (according to Table 2)?
3) If Q2 is a closed switch is the gate of Q3 a high or low voltage?
4) If the gate of Q3, a PMOS FET, is at a low voltage, is Q3 an open or closed switch (according to Table 2)?
5) When Q3 is a closed switch (assume that V_REC+ is 15V DC):
   a. Can current flow through diode D5? Why or why not?
   b. Use the information from the preceding questions to fill in the blanks in the following schematic. Use this schematic to answer the remaining questions.

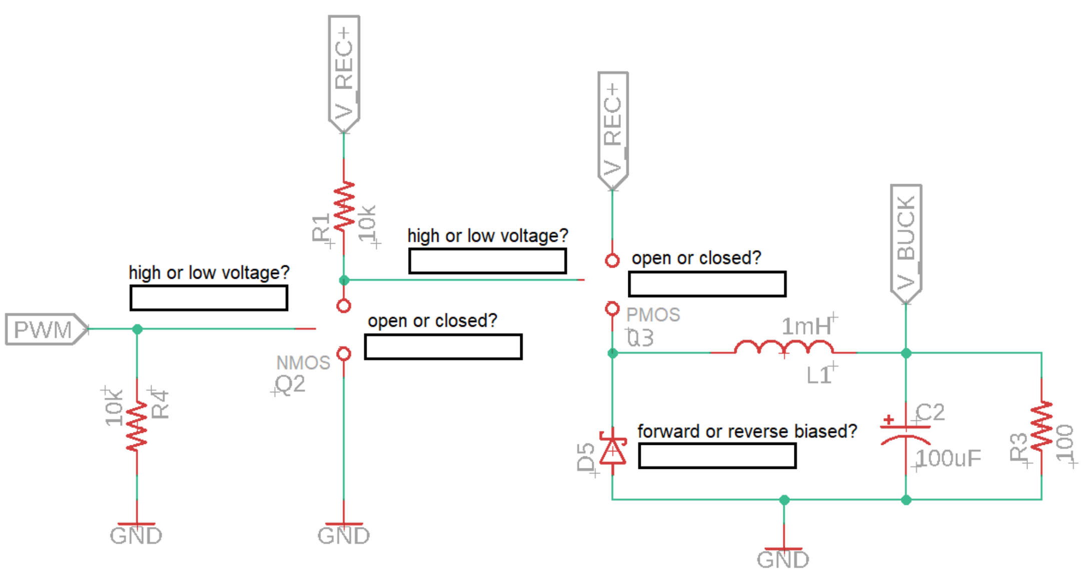


   c. When current flows through inductor L1 does the electric field across the inductor increase or the magnetic field through the inductor increase?

d. After current passes out of inductor L1, where does it flow?
e. When current flows through capacitor C2 what increases? The electric field across the capacitor or the magnetic field through the capacitor?
f. How many ways is energy being stored when the Q3 switch is closed?
g. If switch Q3 was closed for a very long time, how do the inductor and capacitor behave?
    i. Does the inductor act like a short circuit or as a open circuit?
    ii. Does the capacitor act like a short circuit or an open circuit?
h. Assuming that switch Q3 has been closed for a very long time, how much current runs through resistor R3 if V_REC+ is 15V?
i. Would you expect the voltage V_BUCK to increase or decrease while the PWM signal is at a high voltage?

Let's look at the circuit shown in Figure 12 when the **PWM signal is at a low voltage**. Let's start with Q2 because it is directly connected to the PWM signal.

1) When the gate of Q2, a NMOS FET, is at a low voltage, is Q2 an open or closed switch (according to Table 2)?
2) If Q2 is an open switch is the gate of Q3 a high or low voltage?
3) If the gate of Q3, a PMOS FET, is at a high voltage, is Q3 an open or closed switch (according to Table 2)?
4) When Q3 is an open switch:
    a. Use the information from the preceding questions to fill in the blanks in the following schematic. Use this schematic to answer the remaining questions. If you are unsure about the biasing of diode D5, leave this box blank for now.



a. Can current from V_REC+ flow into the inductor? Why or why not?
b. Is any external energy source connected to the components D5, L1, C2, R3?
c. Is there any stored energy in the components D5, L1, C2, R3? If so where?

d. If there is no external energy source connected to D5, L1, C2, R3 is:
   i. The inductor magnetic field building or collapsing?
   ii. The capacitors electric field building or collapsing?
e. The collapsing magnetic field in the inductor causes a current I_L to flow through the inductor in the same direction (left to right) as the charging current. By KCL, what is the current through the diode D5?
f. Is D5 forward or reverse biased? Mark your answer in the figure above if you have not already done so.
g. Write a KVL around the loop formed from the loop D5, L1, C2, R3.
   - Start at the anode of D5
   - Call the voltage drop across the parallel pair C2/R3, V_BUCK
   - Call the voltage drop across the diode V_{diode}
   - Call the voltage drop across the inductor $V_{inductor}$
h. Using your KVL and approximate the voltage drop of the diode as 0V, what is the voltage across the inductor equal to?
i. Is the voltage across the inductor the same or opposite from the case when the PWM signal is at a high voltage?
j. Would you expect the voltage V_BUCK to increase or decrease while the PWM signal is at a low voltage?
k. When V_BUCK is decreasing what is the sign of d V_BUCK/dt (positive or negative)?
l. If d V_BUCK/dt is negative; use the capacitor equation and passive sign convention to determine the flow of current through the capacitor?.
m. What are the sources of current that are feeding the resistor R3?

Buck Converter Simulation:

In order to help you move more quickly through the lab, you can use my Multisim simulation available at [this link] and shown in Figure 13.

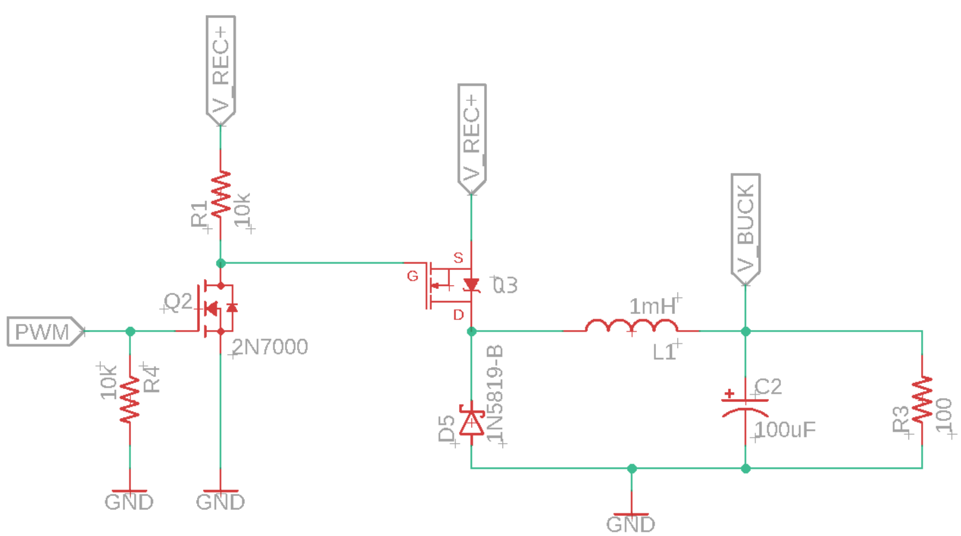


Figure 13: The simulation of the buck converter is available on Multisim. Save yourself some time and use the link provided.

Before running your simulation, please answer the following questions.

1) In the absence of a PWM signal (maybe you haven't attached the function generator to the PWM input or have not enabled the PWM output from the function generator), what voltage is being applied to the base of Q2?
2) When the gate of Q2, a NMOS FET, is at a low voltage, is Q2 an open or closed switch (according to Table 2)?
3) If Q2 is an open switch is the gate of Q3 a high or low voltage?
4) If the gate of Q3, a PMOS FET, is at a high voltage, is Q3 an open or closed switch (according to Table 2)?
5) If Q3 is an open switch is the buck converter receiving any energy from V_RECT+?
6) What is the role of the pair of 10k resistors, R1 and R4, in Figure 13?

Now you are ready to measure the relationship between the load resistance and duty cycle complete Table 3. To do this you will need to change the load and the duty cycle.

- Change the load by changing the resistor value.
- Change the PWM duty cycle by double-click the 5V source and modify the value.

The output voltage will have some ripple. Ignore this ripped when making measurements in the two **average** columns; make your best estimate of the average output voltage at 40ms and write your answer to 2-significant figures. Make two measurements of the ripple voltage at 10% and 90% duty cycle with a 100Ω load.

*Table 3: Using the simulator, record Vout in the **average** columns. Measure ripple voltage for two duty cycles in the **ripple** column.*

	Rload = 100Ω	Rload = 100Ω	Rload = 470Ω
PWM Duty Cycle	V_BUCK (**average**)	V_BUCK (**ripple**)	V_BUCK (**average**)
5%			
10%			
25%			
50%			
75%			
90%			
95%			

Buck Converter Implementation:

Now it's time to run the buck converter from the AC input to the stepped down output. To do this:

1) Connect the KS21239L4 Transformer to the AC inputs on the Buck Converter PCB.
2) Connect a function generator to the PWM and VSS inputs of the Buck Converter PCB.
3) Set the function generator to:
    a. Frequency of 10kHz
    b. Amplitude of 5V, between 0V and 5V

c. Adjust the duty cycle between 5% to 95%

4) Attach channel 1 of the oscilloscope to the V_BUCK terminal and the scope GND clip to GND.

You should have something that looks like Figure 14.

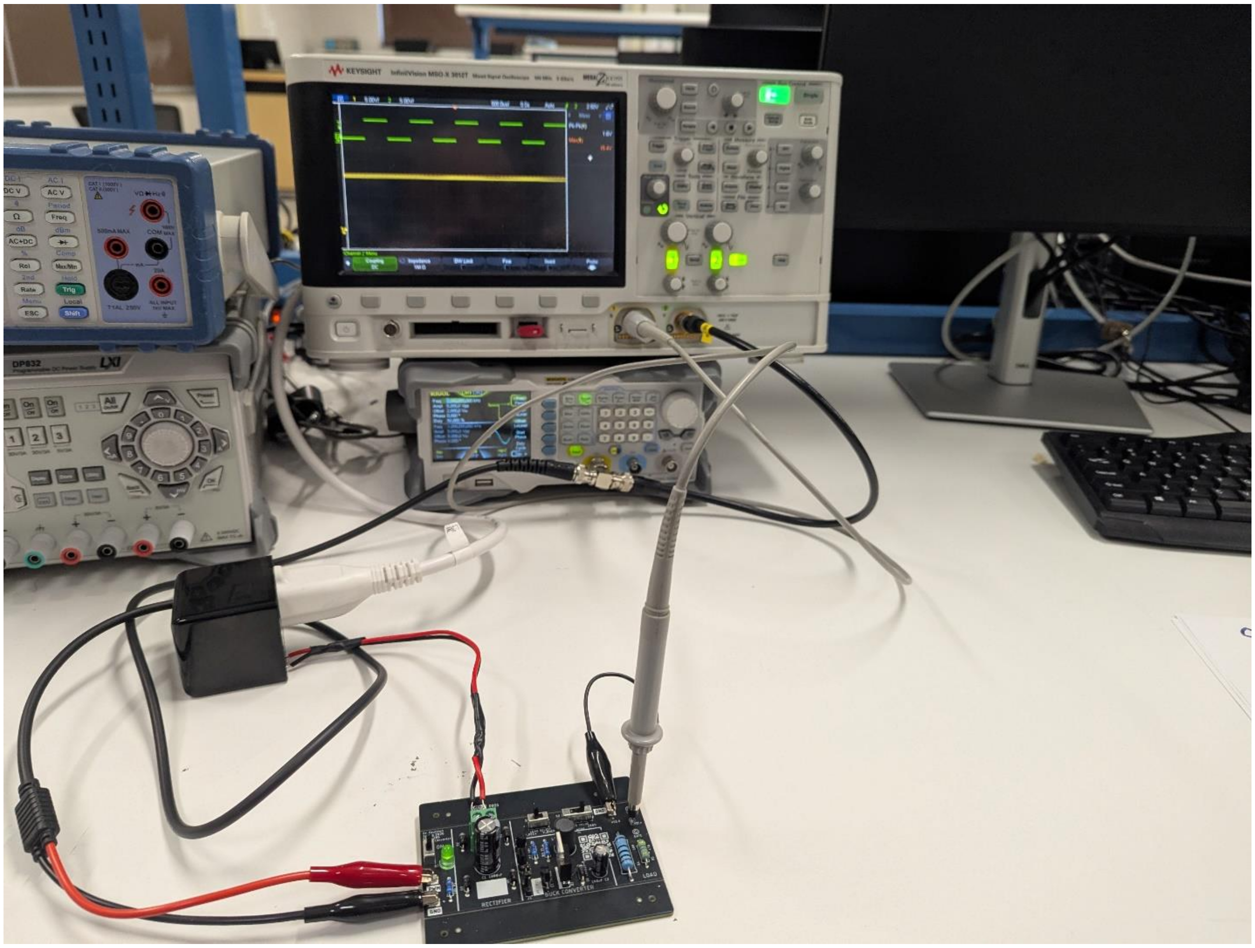


Figure 14: Figure showing the complete setup of the AC input to the Power Converion PCB to loaded output with osciliiscope and function generator

The output voltage will have some ripple. Ignore this ripped when making measurements in the two **average** columns; make your best estimate of the average output voltage and write your answer to 2-significant figures. Make two measurements of the ripple voltage at 10% and 90% duty cycle with a 100Ω load. Record your measurements in Table 4.

*Table 4: Using the Power Conversion PCB, record V_BUCK in the **average** columns. Measure ripple voltage for two duty cycles in the **ripple** column.*

	Rload = 100Ω	Rload = 100Ω	Rload = 470Ω
PWM Duty Cycle	V_BUCK (average)	V_BUCK (ripple)	V_BUCK (average)
5%			
10%			
25%			
50%	15.0		
75%			
90%			16.6V
95%			

Use the data you obtained in Table 3 and Table 4 to answer the following questions. I expect your answer to the question "What is the relationship between X and Y?" to be one of the following

- The larger X gets, the larger Y gets.
- X and Y are unrelated.
- The larger X gets, the smaller Y gets.
- There is no relationship between X and Y.

What is the relationship between the buck convert output and the input parameters?

1. What is the relationship between the load and ripple voltage?
2. What is the relationship between the load and output voltage?
3. What is the relationship between duty cycle and output voltage?
4. What is the relationship between duty cycle and ripple voltage?

Turn in

Make a record of your response to numbered items below and turn them in a single copy as your team's solution on Canvas using the instructions posted there. Include the names of both team members at the top of your solutions. Use complete English sentences to introduce what each of the following listed items (below) is and how it was derived.

Full Bridge Rectifier Theory

Steps 1 – 8

Full Bridge Rectifier Simulation

Table 1

Full Bridge Rectifier Implementation

Steps 1-7

Table 1

Oscilloscope trace with 100Ω and 470Ω load

Buck Converter Theory

PWM High Steps 1-5
PWM Low Steps 1-4

Buck Converter Simulation

Steps 1-6
Table 3

Buck Converter Implementation

Steps 1-4
Complete Table 4
Relationships 1-4